

MATH 3191 Written Homework 4

____/20 points

Name: Key

Due: September 20th

Show all work that leads to your answers. I will be providing feedback on the work as well as the answers!

1. (20 points) Let

$$A = \begin{bmatrix} b & 2 & 3 \\ 3 & 7 & 1 \\ 8 & 5 & 8 \end{bmatrix}$$

- (a) (10 points) Treating the columns of A as separate vectors, determine what value of b will make the columns of A linearly dependent. *Hint:* Consider the definition of linear dependence, you will need to solve a linear system
- (b) (10 points) Use the determinant of A to find a value of b to make A have linearly dependent rows

Solution: Note: I tried to word these parts to ensure each are done differently, and if you asked me I tried to clarify. However if you used the determinant for both that is fine as I am realizing now I was not as clear as I originally thought.

(a) For this, we are wanting to determine a value for b such that

$$c_1 \begin{bmatrix} b \\ 3 \\ 8 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 7 \\ 5 \end{bmatrix} + c_3 \begin{bmatrix} 3 \\ 1 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

has a non-trivial solution. As usual, there are many ways of doing this, but the method I will use is to rewrite above to have the vector with the unknown value b as the “last” vector. That gives us the matrix B as below

$$B = \begin{bmatrix} 2 & 3 & b \\ 7 & 1 & 3 \\ 5 & 8 & 8 \end{bmatrix}$$

The reason that we do this is because I find it easier if we don't have to worry about the value of b during row reduction. In the original matrix A we have to ensure that b is not 0 before dividing by it. This is doable but kind of annoying.

$$\begin{aligned}
\begin{bmatrix} 2 & 3 & b \\ 7 & 1 & 3 \\ 5 & 8 & 8 \end{bmatrix} &\xrightarrow{R_1 \leftarrow \frac{1}{2} R_1} \begin{bmatrix} 1 & \frac{3}{2} & \frac{b}{2} \\ 7 & 1 & 3 \\ 5 & 8 & 8 \end{bmatrix} \\
&\xrightarrow{R_2 \leftarrow R_2 - 7R_1} \begin{bmatrix} 1 & \frac{3}{2} & \frac{b}{2} \\ 0 & -\frac{19}{2} & 3 - \frac{7b}{2} \\ 5 & 8 & 8 \end{bmatrix} \\
&\xrightarrow{R_3 \leftarrow R_3 - 5R_1} \begin{bmatrix} 1 & \frac{3}{2} & \frac{b}{2} \\ 0 & -\frac{19}{2} & 3 - \frac{7b}{2} \\ 0 & \frac{1}{2} & 8 - \frac{5b}{2} \end{bmatrix} \\
&\xrightarrow{\begin{matrix} R_2 \leftarrow -\frac{2}{19} R_2 \\ R_3 \leftarrow 2R_3 \end{matrix}} \begin{bmatrix} 1 & \frac{3}{2} & \frac{b}{2} \\ 0 & 1 & -\frac{6}{19} + \frac{7b}{19} \\ 0 & 1 & 16 - 5b \end{bmatrix} \\
&\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & \frac{3}{2} & \frac{b}{2} \\ 0 & 1 & -\frac{6}{19} + \frac{7b}{19} \\ 0 & 0 & \frac{310}{19} - \frac{102b}{19} \end{bmatrix} \\
&\xrightarrow{R_3 \leftarrow \frac{19}{2} R_3} \begin{bmatrix} 1 & \frac{3}{2} & \frac{b}{2} \\ 0 & 1 & -\frac{6}{19} + \frac{7b}{19} \\ 0 & 0 & 155 - 51b \end{bmatrix}
\end{aligned}$$

Since we want to ensure that this matrix is linearly **dependent**, we want to ensure that $155 - 51b = 0$, which happens when

$$b = \frac{155}{51}.$$

- (b) Recall that for all matrices, $\det(A) = \det(A^\top)$, and a matrix has linearly dependent rows and columns if and only if it is not-invertible, which is true if and only if its determinant is 0. So, we will determine a value b such that the determinant of this matrix is 0.

For this one, we will use the 3×3 determinant formula as discussed in class.

$$\begin{aligned}
\det(A) &= b(7 \cdot 8 - 1 \cdot 5) - 2(3 \cdot 8 - 1 \cdot 8) + 3(3 \cdot 5 - 7 \cdot 8) \\
&= 51b - 32 - 123 \\
&= 51b - 155
\end{aligned}$$

now, we set this equation equal to 0, and solve for b to get

$$b = \frac{155}{51}$$

which was the same as the first part, which is a good sign!